

# Understanding random effects as priors

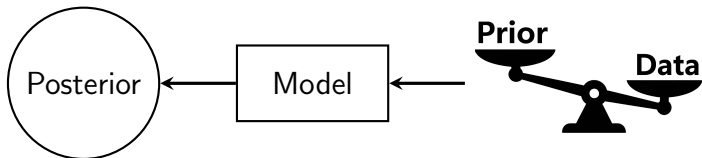
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# What is a prior?

To put it simply:



# Our mathematical model, in full

$$Y \sim \text{Normal}(\mu, \sigma_\epsilon)$$

$$\mu = \alpha + \beta X$$

$$\alpha \sim \text{Normal}(0, 10)$$

$$\beta \sim \text{Normal}(0, 10)$$

$$\sigma_\epsilon \sim \text{Normal}^+(0, 10)$$

Whiteboard:

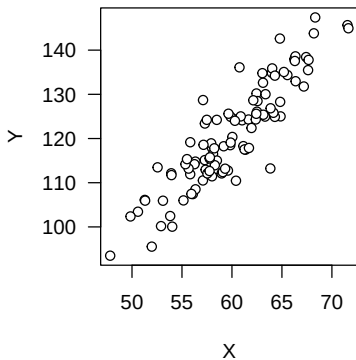
1. Draw prior distributions
2. Update it with strong and weak data

## Five levels of priors

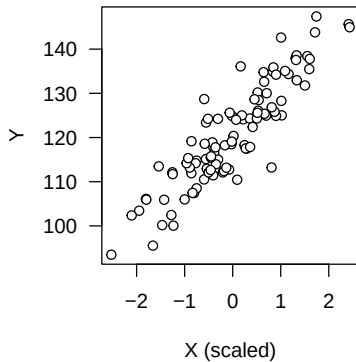
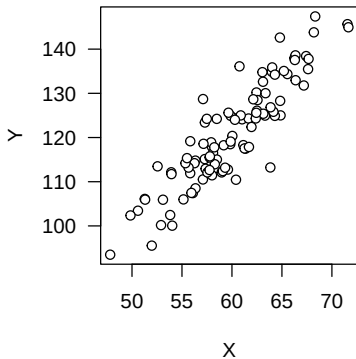
| Prior type                 | Example                      |
|----------------------------|------------------------------|
| Flat                       | $\text{Normal}(0, \infty)$   |
| Super-vague but proper     | $\text{Normal}(0, 10^6)$     |
| Weakly informative         | $\text{Normal}(0, 10)$       |
| Generic weakly informative | $\text{Normal}(0, 1)$        |
| Informative                | $\text{Normal}(-0.47, 0.15)$ |

Whiteboard: overlay priors

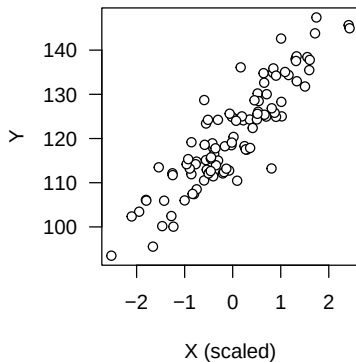
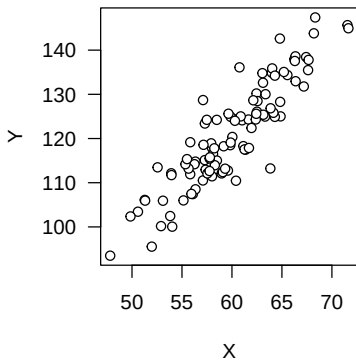
# Prior guessing



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Prior choice depends on the data scale!

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- Think of “subjectivity” as “transparency”

## From fixed to random effects: it's in the priors

$$Y \sim \text{Normal}(\mu, \sigma_\epsilon)$$

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